

Quadratic Forms (二次型)

Quadratic Forms

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Quadratic Forms

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Introduction

A quadratic equation in n variables $x_1, x_2, \dots, x_n$ is one of the form

$$\mathbf{x}^T A \mathbf{x} + B \mathbf{x} + \alpha = 0.$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$, A is an $n \times n$ real symmetric matrix, B is a $1 \times n$ matrix, and α is a scalar. The vector function

$$f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \left(\sum_{j=1}^n a_{ij} x_j \right) x_i$$

is the quadratic form in n variables associated with the quadratic equation.

Three Unknowns

In the case of three unknowns, if

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, A = \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix}, B = \begin{bmatrix} g & h & i \end{bmatrix},$$

then the quadratic equation is

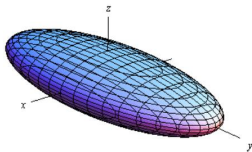
$$ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz + gx + hy + iz + \alpha = 0.$$

The graph of a quadratic equation in three variables is called a quadric surface (二次曲面).

Ellipsoids (椭球面)

1. Ellipsoids

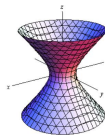
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad a > 0, \quad b > 0, \quad c > 0.$$



Hyperboloid of One Sheet(单叶双曲面)

2. Hyperboloid of One Sheet:

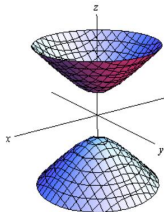
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \quad (a > 0, b > 0, c > 0).$$



Hyperboloid of Two Sheets(双叶双曲面)

3. Hyperboloid of Two Sheets:

$$\frac{z^2}{a^2} - \frac{x^2}{b^2} - \frac{y^2}{c^2} = 1, \quad a, b, c > 0.$$

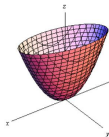


Elliptic Paraboloid(椭圆抛物面)

4. Elliptic Paraboloid:

$$\frac{x^2}{2p} + \frac{y^2}{2q} = z,$$

where p, q have the same sign.

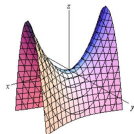


Hyperbolic Paraboloid(双曲抛物面)

5. Hyperbolic Paraboloid:

$$\frac{x^2}{2p} - \frac{y^2}{2q} = z,$$

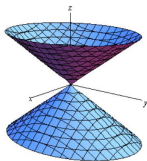
where p, q have the same sign.



Cones (圆锥面)

6. Cones:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0, \quad a, b, c > 0.$$



Example 1

Example

Given the quadratic equation

$$2x_1x_2 + 2x_1x_3 - 6x_2x_3 = 1.$$

Find a change of coordinates so that the resulting equation represents a quadric surface in standard position.

The left hand side is of a quadratic form, and its matrix is

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -3 \\ 1 & -3 & 0 \end{bmatrix}.$$

Approach 1: Completing the Squares (part 1)

Let

$$\begin{cases} x_1 = y_1 + y_2, \\ x_2 = y_1 - y_2, \\ x_3 = y_3, \end{cases}$$

then

$$f(x_1, x_2, x_3) = 2(y_1 - y_3)^2 - 2y_3^2 - 2y_2^2 + 8y_2y_3.$$

Let

$$\begin{cases} z_1 = y_1 - y_3, \\ z_2 = y_2, \\ z_3 = y_3, \end{cases} \Leftrightarrow \begin{cases} y_1 = z_1 + z_3, \\ y_2 = z_2, \\ y_3 = z_3, \end{cases}$$

Approach 1: Completing the Squares (part 2)

It follows that

$$f(x_1, x_2, x_3) = 2z_1^2 - 2(z_2 - 2z_3)^2 + 6z_3^2.$$

Let

$$\begin{cases} w_1 = z_1, \\ w_2 = z_2 - 2z_3, \\ w_3 = z_3, \end{cases} \Leftrightarrow \begin{cases} z_1 = w_1, \\ z_2 = w_2 + 2w_3, \\ z_3 = w_3, \end{cases}$$

then

$$f(x_1, x_2, x_3) = 2w_1^2 - 2w_2^2 + 6w_3^2.$$

Approach 1: Completing the Squares (part 3)

The change of coordinates is give by

$$\begin{aligned}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} .\end{aligned}$$

The quadratic equation becomes $f(x_1, x_2, x_3) = 2w_1^2 - 2w_2^2 + 6w_3^2 = 1$.

Therefore it represents a hyperboloid of one sheet in standard position.

Approach 2: Using Matrix Multiplication (part 1)

Let

$$C_1 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

then

$$\begin{aligned} A_1 = C_1^T A C_1 &= \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -3 \\ 1 & -3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & -2 \\ 0 & -2 & 4 \\ -2 & 4 & 0 \end{bmatrix}. \end{aligned}$$

Approach 2: Using Matrix Multiplication (part 2)

Let

$$C_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

then

$$\begin{aligned} A_2 = C_2^T A_1 C_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -2 \\ 0 & -2 & 4 \\ -2 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 4 \\ 0 & 4 & -2 \end{bmatrix}. \end{aligned}$$

Approach 2: Using Matrix Multiplication (part 3)

Let

$$C_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix},$$

then

$$\begin{aligned} A_3 = C_3^T A_2 C_3 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 4 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 6 \end{bmatrix}. \end{aligned}$$

Approach 2: Using Matrix Multiplication (part 4)

A_3 is already diagonal, therefore if we let

$$\begin{aligned} C = C_1 C_2 C_3 &= \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix}, \end{aligned}$$

then

$$C^T A C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 6 \end{bmatrix}.$$

Approach 2: Using Matrix Multiplication (part 5)

If we choose

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix},$$

then the quadratic equation in this example becomes

$$2y_1^2 - 2y_2^2 + 6y_3^2 = 1.$$

Therefore the equation represents a hyperboloid of one sheet in standard position.

Approach 3: Spectral Theorem

Since A is a symmetric matrix, therefore by the spectral theorem, we can find an orthogonal matrix Q such that $Q^{-1}AQ = \Lambda$. In particular

$$A = Q\Lambda Q^T = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \begin{bmatrix} q_1^T \\ q_2^T \\ q_3^T \end{bmatrix}.$$

If we let $y = Q^T x$, then

$$x^T A x = x^T Q \Lambda Q^T x = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2.$$

The eigenvalues of A are $3, \frac{-3+\sqrt{17}}{2}, \frac{-3-\sqrt{17}}{2}$. Therefore the equation represents a hyperboloid of one sheet in standard position.

Exercises

1. 设二次型 $f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + 4x_1x_2 + 4x_1x_3 + 4x_2x_3$, 则 $f(x_1, x_2, x_3) = 2$ 在空间直角坐标系下表示的二次曲面为
- (A) 单叶双曲面
(B) 双叶双曲面
(C) 椭球面
(D) 柱面
2. 已知二次曲面方程 $x^2 + ay^2 + z^2 + 2bxy + 2xz + 2yz = 4$ 可以经过正交变换

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = P \begin{bmatrix} \xi \\ \eta \\ \zeta \end{bmatrix}$$

化为椭圆柱面方程 $\eta^2 + 4\xi^2 = 4$, 求 a, b 的值和正交矩阵 P .

Exercises

3. 已知二次型 $f(x_1, x_2, x_3) = x^T A x$ 在正交变换 $x = Qy$ 下的标准形为 $y_1^2 + y_2^2$, 且 Q 的第三列为 $(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2})^T$.

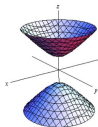
(I) 求矩阵 A .

(II) 证明 $A + E$ 为正定矩阵, 其中 E 为三阶单位矩阵.

4. 设 A 为三阶实对称矩阵, 如果二次曲面方程

$$\begin{bmatrix} x & y & z \end{bmatrix} A \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

在正交变换下的标准方程的图形如图所示,



则 A 的正特征值的个数为 _____.

Exercises

5. 已知二次型

$$f(x_1, x_2, x_3) = (1-a)x_1^2 + (1-a)x_2^2 + 2x_3^2 + 2(1+a)x_1x_2$$

的秩为 2.

- (I) 求 a 的值;
- (II) 求正交变换 $x = Qy$, 把 $f(x_1, x_2, x_3)$ 化为标准形;
- (III) 求方程 $f(x_1, x_2, x_3) = 0$ 的解.

6. 设二次型

$$f(x_1, x_2, x_3) = ax_1^2 + ax_2^2 + (a-1)x_3^2 + 2x_1x_3 - 2x_2x_3.$$

- (I) 求二次型 f 的矩阵的所有特征值;
- (II) 若二次型 f 的规范形为 $y_1^2 + y_2^2$, 求 a 的值.